

CORE JAVA

- 1) class ,object , JVM Architecture (in detail)
- 2) static context (static variables /methods/blocks)
- 3) non-static context(instance variables/methods /blocks)
- 2) OOPS -> 1) encapsulation
 - 2) inheritance (IS -A relation)
 - 3) Polymorphism
 - 4) Abstraction
- 3) Has -A relation
- 4) Aggregation/Composition
- 5) inner class /anonymous inner class /static inner class / Non-static inner class
- 6) Arrays / Strings
- 7) Exception handling (try, catch, finally, try-with-resources, throw, throws, checked/unchecked Exception, custom Exception)
- 8) Multithreading -> 1) thread life cycle
 - 2) methods -> sleep(),join(),yield(),interrupt()
 - 3) Synchronization (object level & class level)
 - 4) inter-thread-communication(ITC)--> wait(),notify(), notifyAll()
 - 5) deadlock

6) Daemon thread

7) Thread pool , Executor framework

9) collections

->List

->Vector

->Stack

->ArrayList

->LinkedList

->Set

->HashSet

->LinkedHashSet

->SortedSet

->NavigableSet

->TreeSet

->Queue

->PriorityQueue

->Deque

->ArrayDeque

MAP

->HashMap

->LinkedHashMap

->SortedMap

->NavigableMap

->TreeMap

->Hashtable

->WeakHashMap

->IdentityHashMap

Enumeration , list of ways(8) to iterate the collections, properties class, list iterators
,Spliterator ,fail-fast/fail-safe iterators

concurrent collections(CopyOnWriteArrayList,ConcurrentHashMap)
,ConcurrentModificationException

10)Functional Programming

- 1)functional interfaces
- 2)lambda expression
- 3)method reference

11) Stream API (All the methods in Stream API which is used for logical problems)

12) JAVA 8 futures

13) java utility class (Like Arrays/Collections /collectors)

SPRING

Spring Core:-

- 1) Inversion of Control (IoC)
- 2) Spring Container (BeanFactory,ApplicationContext)
- 3) Spring Bean(Bea definition ,Bean scope (singleton, prototype, request, session, websocket ,application)
- 4) Bean Lifecycle(init-method / destroy-method ,@PostConstruct / @PreDestroy)
- 5) Configuration Types(XML configuration,Java-based configuration,Annotation-based configuration)
- 6) Dependency Injection (DI)
 - >Constructor Injection
 - >Setter Injection
 - >Field Injection
- 7) @Autowired,@Qualifier,@Primary,@Value,@PropertySource,@Profile

Spring Boot:-

- 1)What is Spring Boot

2)Spring Boot vs Spring

3) @SpringBootApplication

4)Auto-configuration

5)Starter dependencies

6) application.properties / application.yml/yaml

7)Embedded servers (Tomcat, Jetty)

8) CommandLineRunner / ApplicationRunner

9) Externalized configuration

Spring MVC / REST:-

1) MVC Architecture /Rest Architecture

2) @Controller vs @RestController

3)@Component,@configuration ,@Controller, @Service, @Repository, @Entity

4) @RequestMapping,@GetMapping, @PostMapping, @PutMapping, @PatchMapping, @DeleteMapping,

5) @RequestParam, @PathVariable ,@RequestBody ,@ResponseBody, @ResponseStatus
(explain all status codes)

6) @ExceptionHandler ,@ControllerAdvice ,@RestControllerAdvice (Global Exception Handling)

7) Validation (@Valid,@NotNull, @Size, etc.)

8) Spring Data JPA -What is JPA ,Hibernate vs JPA ,Repository Pattern,CrudRepository,JpaRepository ,@Query,@Param (JPQL & Native Query) ,Pagination & Sorting

Relationships (OneToOne,OneToMany,ManyToOne,ManyToMany) ,Fetch types (EAGER vs LAZY),Cascade types,Transaction Management(@Transactional),

Optimistic Locking,@Version

Spring Scheduling :-

1)@EnableScheduling ,@Scheduled annotation ,fixedRate,fixedDelay ,initialDelay

2) Cron expressions(Cron format (6 fields))

Spring Redis Cache:-

1) Redis overview

2) Spring Cache abstraction

3)@EnableCaching,CacheManager,RedisCacheManager(@Cacheable,@CachePut,@CacheEvict)

4) Redis connection factory (Lettuce / Jedis)

5)Cache vs DB performance considerations

Microservices with Spring Cloud:-

1) Microservices architecture (RestTemplate ,webclient /Feign Client) (how one service communicating other service overview)

2) Spring Cloud overview(Service Discovery,Eureka Server,Eureka Client,API Gateway,Spring Cloud Gateway,Load Balancing,Spring Cloud LoadBalancer, Configuration Server(Spring Cloud Config),Circuit Breaker(Resilience4j),Distributed Tracing(Sleuth / Zipkin)

3) Centralized Logging(ELK stack)

Spring Security:-

1) What is Spring Security? (Authentication vs Authorization)

2) Security filters

3)In-memory authentication /Database authentication /UserDetails & UserDetailsService ,PasswordEncoder ,Role-based access

4) JWT Authentication

5) OAuth2 (Basic understanding)

6) CSRF / CORS with Security